

Section : Biology

Q.1 Sickle-cell anemia is caused due to:

1. Change of amino acid in β -chain of haemoglobin
2. Change of amino acid in alpha-chain of haemoglobin
3. Change of amino acid in both alpha and β -chains of haemoglobin
4. Change of amino acid in one of the alpha or β -chains of haemoglobin

Options 1. 1

2. 2
3. 3
4. 4

Q.2 The rate of biomass production is called productivity and it is expressed in terms of _____ to compare the productivity of different ecosystems.

1. $\text{gm}^{-2} \text{yr}^{-2}$ or $(\text{kcal m}^{-2}) \text{yr}^{-2}$
2. $\text{gm}^{-2} \text{yr}^{-1}$ or $(\text{kcal m}^{-2}) \text{yr}^{-1}$
3. $\text{gm}^{-1} \text{yr}^{-1}$ or $(\text{kcal m}^{-2}) \text{yr}^{-2}$
4. $\text{gm}^{-1} \text{yr}^{-2}$ or $(\text{kcal m}^{-1}) \text{yr}^{-2}$

Options 1. 1

2. 2
3. 3
4. 4

Q.3 According to the IUCN, the total number of plant and animal species is about :-

1. 1.5 million
2. 2 million
3. 7 million
4. 5 million

Options 1. 1

2. 2
3. 3
4. 4

Q.4 Continued self-pollination results in:

1. Pollen-pistil interaction
2. Inbreeding depression
3. Artificial hybridization
4. Emasculation

Options 1. 1

2. 2
3. 3
4. 4

Q.5 Arrange the following groups of plants in sequential order of their occurrence?

- (A) Conifers
- (B) Progymnosperms
- (C) Ginkgos
- (D) Angiosperms

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (B), (A), (C), (D)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.6 In which type of RNA, methyl guanosine triphosphate capping is added to 5'-end ?

1. rRNA
2. mRNA
3. tRNA
4. hnRNA

Options 1. 1

2. 2
3. 3
4. 4

Q.7 The characteristics of Baculoviruses are:

- (A) They are used as biological control agents.
- (B) They are excellent candidates for species-specific, narrow spectrum insecticidal application.
- (C) They are excellent candidates for species-specific, broad spectrum insecticidal application.
- (D) No negative impacts on plants, animals, birds, fishes and other non-targeted insects.

Choose the correct answer from the options given below:

1. (A), (C) and (D) only
2. (A), (B) and (D) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.8 Which of the following combination of hormones are secreted during pregnancy in humans?

- (A) hCG
- (B) LH
- (C) hPL
- (D) Relaxin

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (B), (C), and (D) only
3. (A), (C) and (D) only
4. (A), (B) and (C) only

Options 1. 1

2. 2
3. 3
4. 4

Q.9 Which organism is not an example of recent extinction?

1. Dodo
2. Bengal tiger
3. Javan tiger
4. Passenger pigeon

Options 1. 1

2. 2
3. 3
4. 4

Q.10 Which one of the followings is a hormone releasing IUD?

1. LNG-20
2. Lippes Loop
3. Cu7
4. Multiload 375

Options 1. 1

2. 2
3. 3
4. 4

Q.11 In grass families, the cotyledon is called as:

1. Coleoptile
2. Coleorhiza
3. Thalamus
4. Scutellum

Options 1. 1

2. 2
3. 3
4. 4

Q.12 Arrange the given steps for regulation of gene expression in correct sequence?

- (A) Regulation of splicing
- (B) Transport of mRNA from the nucleus to the cytoplasm
- (C) Translation
- (D) Formation of primary transcript (Transcription)

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (B), (A), (C), (D)
3. (D), (A), (B), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.13 The distance between a base pair (bp) in a DNA helix is approximately:

1. 0.14 nm.
2. 0.64 nm
3. 0.34 nm
4. 0.74 nm

Options 1. 1

2. 2
3. 3
4. 4

Q.14 Polygenic traits are controlled by:

1. One gene
2. Two genes
3. Three or more genes
4. One or two genes

Options 1. 1

2. 2
3. 3
4. 4

Q.15 In your opinion, which is the most effective way to conserve the plant diversity of an area?

1. Tissue culture method
2. Creation of a biosphere reserve
3. Creation of a botanical garden
4. Development of a seed bank

Options 1. 1

2. 2
3. 3
4. 4

Q.16 Match List-I with List-II

List-I

Strategies of an organism

(A) Organisms breed only once in their lifetime

(B) Organisms breed many times during their lifetime

List-II

Example

(I) Desert lizard

(II) Oysters

(C) Organisms produce large number of small sized offspring (III) Salmon fish and bamboo

(D) Some organisms show behavioral responses with variation in their environment (IV) Birds and mammals

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

2. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)

4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.17 Human insulin was prepared by:

1. Herbert Boyer

2. M.S. Swaminathan

3. Eli Lilly

4. James Watson

Options 1. 1

2. 2

3. 3

4. 4

Q.18 Match List-I with List-II

List-I

List-II

Contraceptive methods

Respective actions

(A) Diaphragms

(I) Inhibit ovulation and implantation

(B) Contraceptive pills

(II) Increase phagocytosis of sperm within uterus

(C) Intra Uterine Devices

(III) Absence of menstrual cycle and ovulation following parturition

(D) Lactational amenorrhea

(IV) They cover the cervix blocking the entry of sperms

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

2. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

3. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)

4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.19 *Methanobacterium* is also present in the _____ of cattle which helps in breakdown of cellulose.

1. Intestine

2. Brain

3. Stomach

4. Lungs

Options 1. 1

2. 2

3. 3

4. 4

Q.20 Choose the correct statements about organic farming:

(A) It does not use pesticides and urea.

(B) It uses only naturally produced inputs like compost.

(C) It utilizes genetically modified crops.

(D) It produces fruits rich in minerals and vitamins.

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (A), (B) and (D) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.21 Which body of the Government of India regulates GM research and safety of introducing GM organisms for public services?

1. Genetic Engineering Authority Committee
2. Research Committee on Genetic Manipulation
3. Bio-safety committee
4. Genetic Engineering Approval Committee

Options 1. 1

2. 2
3. 3
4. 4

Q.22 Arrange the given species in correct trophic levels in a grassland ecosystem?

- (A) Crow
- (B) Vulture
- (C) Rabbit
- (D) Grass

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (D), (C), (A), (B)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.23 Identify the correct sequence of structures formed during spermatogenesis in humans?

- (A) Spermatozoa
- (B) Secondary spermatocytes
- (C) Spermatogonia
- (D) Spermatid

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (D), (C), (B), (A)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.24 Somaclones can be defined as:

1. They have the capacity to generate a whole plant from any explant.
2. They are genetically identical to the original plant.
3. They are genetically modified pest resistant plants.
4. They are the plants, infected with virus.

Options 1. 1

2. 2
3. 3
4. 4

Q.25 In *Antirrhinum* (Snapdragon), a red flower was crossed with a white flower and in F_1 generation all pink flowers were obtained. When pink flowers were self crossed, F_2 generation showed white, red and pink flowers.

Choose the correct statements for this case:

- (A) This experiment does not follow the Principle of Dominance.
- (B) Pink colour in F_1 is due to incomplete dominance.
- (C) Ratio of F_2 is 1 (red): 2 (pink): 1 (white).
- (D) Law of segregation does not apply in this case.

Choose the correct answer from the options given below:

- 1. (A), (B) and (D) only
- 2. (A), (B) and (C) only
- 3. (A), (B), (C) and (D)
- 4. (B), (C) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.26 Arrange the given steps of decomposition cycle in a terrestrial ecosystem in correct sequence?

- (A) Bacterial and fungal enzymes degrade detritus into simpler inorganic substances and its catabolism.
- (B) Earthworms break-down detritus into smaller particles and is called fragmentation.
- (C) Leaching - water soluble inorganic nutrients go down into the soil horizon.
- (D) Microbes release inorganic nutrients in the soil and it is called mineralisation.

Choose the correct answer from the options given below:

- 1. (A), (B), (C), (D)
- 2. (B), (C), (A), (D)
- 3. (B), (A), (D), (C)
- 4. (C), (B), (D), (A)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.27 Match List-I with List-II

List-I

List-II

Products

Microorganisms

- (A) Wine or beer (I) *Penicillium notatum*
- (B) Swiss Cheese (II) *Aspergillus niger*
- (C) Antibiotic (III) *Propionibacterium sharmanii*
- (D) Citric Acid (IV) *Sachharomyces cerevisiae*

Choose the correct answer from the options given below:

- 1. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)
- 2. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)
- 3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
- 4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.28 Match List-I with List-II

List-I

List-II

Structures

Quantity/Dimensions

- (A) Length of ovary (I) 250
- (B) Number of mammary lobes (II) 4 to 5 cm
- (C) Number of compartment in testis (III) 2 to 4 cm

(D) Length of testis

(IV) 15 to 20

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.29 Which one of the following is not a method of in-situ conservation of biodiversity?

1. Sacred grove
2. Biosphere reserve
3. Wildlife sanctuary
4. Botanical garden

Options 1. 1

2. 2
3. 3
4. 4

Q.30 The function of germ pore in a pollen grain is:

1. To protect the pollen grain
2. To form the pollen tube
3. To divide the cells of pollen grain
4. To store the food

Options 1. 1

2. 2
3. 3
4. 4

Q.31 Assisted reproductive technology, IVF involves transfer of:

1. Ovum into the fallopian tube
2. Zygote into the fallopian tube
3. Zygote into the uterus
4. Embryo with 16 blastomeres into the fallopian tube

Options 1. 1

2. 2
3. 3
4. 4

Q.32 Who proposed the "Embryological support for Evolution"?

1. Charles Darwin
2. Ernst Heckel
3. Hugo De Vries
4. Alfred Wallace

Options 1. 1

2. 2
3. 3
4. 4

Q.33 Which one of the followings is not a feature of insecticidal protein crystals formed by *Bacillus thuringiensis*: -

1. When formed they exist as inactive protoxin.
2. They are converted into an active form of toxin in the alkaline pH of the gut, which solubilize the crystals.
3. They are effective against certain insects such as *lepidopterans*, *coleopterans* and *dipterans*.
4. The activated toxin does not bind the surface of midgut epithelial cells which leads to pore formation, cell swelling and lysis.

Options 1. 1

2. 2
3. 3
4. 4

Q.34 The seminal plasma is rich in:

- (A) Fructose
- (B) Amino acids
- (C) Calcium
- (D) Certain enzymes

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (C) and (D) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.35 If in a pond there were 20 lotus plants last year and through reproduction 8 new plants are added, taking the current population to 28, the birth rate would be:

1. 0.4 offspring per lotus per year
2. 0.8 offspring per lotus per year
3. 0.5 offspring per lotus per year
4. 0.2 offspring per lotus per year

Options 1. 1

2. 2
3. 3
4. 4

Q.36 De Vries gave his mutation theory on organic evolution while working on-

1. *Pisum sativum*
2. *Drosophila melanogaster*
3. Evening primrose
4. *Althea rosea*

Options 1. 1

2. 2
3. 3
4. 4

Q.37 Which one of the followings is not a transgenic animal?

1. Sheep
2. Fish
3. Lion
4. Rabbit

Options 1. 1

2. 2
3. 3
4. 4

Q.38 The polluting potential of waste water is determined by:

1. Biochemical oxygen demand
2. Presence of sludge
3. Biogas production
4. Flocs formation

Options 1. 1

2. 2
3. 3
4. 4

Q.39 Silencing of mRNA has been used in producing transgenic plants which are resistant to:

1. Boll worms
2. Nematodes
3. White rusts
4. Bacterial blights

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.40 Match List-I with List-II

List-I	List-II
Types of parasite	Examples
(A) Ectoparasite	(I) Cuckoo
(B) Brood parasite	(II) Human liver fluke
(C) Trematode parasite	(III) Mosquito
(D) Malarial parasite	(IV) Lice on human

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)
3. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.41 Read the following paragraph and answer the given questions:

DRUGS AND ALCOHOL ABUSE

The drugs, commonly abused are opioids, cannabinoids, and coca alkaloids. Opioids are the drugs, which bind to specific opioid receptors present in our central nervous system and gastrointestinal tract. Heroin, commonly called smack is chemically diacetylmorphine which is a white, odourless, bitter crystalline compound. Cannabinoids are a group of chemicals, which interact with cannabinoid receptors present principally in the brain. Natural cannabinoids are obtained from the inflorescences of the plant *Cannabis sativa*. The flower tops, leaves and the resin of cannabis plant are used in various combinations to produce marijuana, hashish, charas and ganja. Generally taken by inhalation and oral ingestion, these are known for their effects on cardiovascular system of the body. It interferes with the transport of the neuro-transmitter dopamine. Cocaine, commonly called coke or crack is usually snorted. It has a potent stimulating action on central nervous system, producing a sense of euphoria and increased energy. Excessive dosage of cocaine causes hallucinations. Other well-known plants with hallucinogenic properties are *Atropa belladonna* and *Datura*. These days cannabinoids are also being abused by some sportspersons. Drugs like barbiturates, amphetamines, benzodiazepines, and other similar drugs, that are normally used as medicines to help patients cope with mental illnesses like depression and insomnia, are often abused. Morphine is a very effective sedative and painkiller, and is very useful in patients who have undergone surgery. Several plants, fruits and seeds having hallucinogenic properties have been used for hundreds of years in folk-medicine, religious ceremonies and rituals all over the globe.

_____ are used as medicines to help patients cope with mental illnesses like depression and insomnia.

1. Amphetamines
2. Cocaine
3. Marijuana
4. Heroin

- Options
1. 1
 2. 2
 3. 3

Q.42 Read the following paragraph and answer the given questions:

DRUGS AND ALCOHOL ABUSE

The drugs, commonly abused are opioids, cannabinoids, and coca alkaloids. Opioids are the drugs, which bind to specific opioid receptors present in our central nervous system and gastrointestinal tract. Heroin, commonly called smack is chemically diacetylmorphine which is a white, odourless, bitter crystalline compound. Cannabinoids are a group of chemicals, which interact with cannabinoid receptors present principally in the brain. Natural cannabinoids are obtained from the inflorescences of the plant *Cannabis sativa*. The flower tops, leaves and the resin of cannabis plant are used in various combinations to produce marijuana, hashish, charas and ganja. Generally taken by inhalation and oral ingestion, these are known for their effects on cardiovascular system of the body. It interferes with the transport of the neuro-transmitter dopamine. Cocaine, commonly called coke or crack is usually snorted. It has a potent stimulating action on central nervous system, producing a sense of euphoria and increased energy. Excessive dosage of cocaine causes hallucinations. Other well-known plants with hallucinogenic properties are *Atropa belladonna* and *Datura*. These days cannabinoids are also being abused by some sportspersons. Drugs like barbiturates, amphetamines, benzodiazepines, and other similar drugs, that are normally used as medicines to help patients cope with mental illnesses like depression and insomnia, are often abused. Morphine is a very effective sedative and painkiller, and is very useful in patients who have undergone surgery. Several plants, fruits and seeds having hallucinogenic properties have been used for hundreds of years in folk-medicine, religious ceremonies and rituals all over the globe.

Which one of the following drugs is known for its effects on the cardiovascular system of the body?

1. Opioids
2. Cocaine
3. Cannabinoids
4. Barbiturates

Options 1. 1

2. 2

3. 3

4. 4

Q.43 Read the following paragraph and answer the given questions:

DRUGS AND ALCOHOL ABUSE

The drugs, commonly abused are opioids, cannabinoids, and coca alkaloids. Opioids are the drugs, which bind to specific opioid receptors present in our central nervous system and gastrointestinal tract. Heroin, commonly called smack is chemically diacetylmorphine which is a white, odourless, bitter crystalline compound. Cannabinoids are a group of chemicals, which interact with cannabinoid receptors present principally in the brain. Natural cannabinoids are obtained from the inflorescences of the plant *Cannabis sativa*. The flower tops, leaves and the resin of cannabis plant are used in various combinations to produce marijuana, hashish, charas and ganja. Generally taken by inhalation and oral ingestion, these are known for their effects on cardiovascular system of the body. It interferes with the transport of the neuro-transmitter dopamine. Cocaine, commonly called coke or crack is usually snorted. It has a potent stimulating action on central nervous system, producing a sense of euphoria and increased energy. Excessive dosage of cocaine causes hallucinations. Other well-known plants with hallucinogenic properties are *Atropa belladonna* and *Datura*. These days cannabinoids are also being abused by some sportspersons. Drugs like barbiturates, amphetamines, benzodiazepines, and other similar drugs, that are normally used as medicines to help patients cope with mental illnesses like depression and insomnia, are often abused. Morphine is a very effective sedative and painkiller, and is very useful in patients who have undergone surgery. Several plants, fruits and seeds having hallucinogenic properties have been used for hundreds of years in folk-medicine, religious ceremonies and rituals all over the globe.

'Coca alkaloids' are obtained from the plant-

1. *Atropa belladonna*
2. *Papaver somniferum*
3. *Cannabis sativa*

4. *Erythroxylum coca*

Options 1. 1

2. 2

3. 3

4. 4

Q.44 Read the following paragraph and answer the given questions:

DRUGS AND ALCOHOL ABUSE

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Which is used as an effective sedative and painkiller and is useful to patients who have undergone surgery?

1. Hashish

2. Morphine

3. Heroin

4. Ganja

Options 1. 1

2. 2

3. 3

4. 4

Q.45 Read the following paragraph and answer the given questions:

DRUGS AND ALCOHOL ABUSE

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been used for hundreds of years in folk-medicine, religious ceremonies and rituals all over the globe.

Heroin (Smack) which is extracted from the latex of poppy plant is chemically called:

1. Dimethylmorphine
2. Diacetylmorphine
3. Acetylmorphine
4. Methylmorphine

Options 1. 1

2. 2

3. 3

4. 4

Q.46 Read the following paragraph and answer the given questions:

In order to link the alien DNA, the vector needs to have very few, preferably single, recognition sites for the commonly used restriction enzymes. Presence of more than one recognition site in the vector will generate several fragments, which will complicate the gene cloning. In order to link the alien DNA, the vector needs to have very few, preferably single, recognition sites for the commonly used restriction enzymes. Presence of more than one recognition sites within the vector will generate several fragments, which will complicate the gene cloning. The ligation of alien DNA is carried out at a restriction site present in one of the two antibiotic resistance genes. The recombinant plasmids will lose tetracycline resistance due to insertion of foreign DNA but can still be selected out from non-recombinant ones by plating the transformants on tetracycline containing medium.

The recombinants will grow in ampicillin containing medium but not on that containing tetracycline. But, non-recombinants will grow on the medium containing both the antibiotics. In this case, one antibiotic resistance gene helps in selecting the transformants, whereas the other antibiotic resistance gene gets 'inactivated due to insertion' of alien DNA, and helps in selection of recombinants. Origin of replication (ori) is a sequence from where replication starts and any piece of DNA when linked to this sequence can be made to replicate within the host cells. This sequence is also responsible for controlling the copy number of the linked DNA. So, if one wants to recover many copies of the target DNA it should be cloned in a vector whose origin support has high copy number.

Multiple recognition sites in vector generate several fragments, which will complicate the-

1. Electrophoresis
2. Recombinant DNA technology
3. Mutation
4. Gene cloning

Options 1. 1

2. 2

3. 3

4. 4

Q.47 Read the following paragraph and answer the given questions:

In order to link the alien DNA, the vector needs to have very few, preferably single, recognition sites for the commonly used restriction enzymes. Presence of more than one recognition site in the vector will generate several fragments, which will complicate the gene cloning. In order to link the alien DNA, the vector needs to have very few, preferably single, recognition sites for the commonly used restriction enzymes. Presence of more than one recognition sites within the vector will generate several fragments, which will complicate the gene cloning. The ligation of alien DNA is carried out at a restriction site present in one of the two antibiotic resistance genes. The recombinant plasmids will lose tetracycline resistance due to insertion of foreign DNA but can still be selected out from non-recombinant ones by plating the transformants on tetracycline containing medium.

The recombinants will grow in ampicillin containing medium but not on that containing tetracycline. But, non-recombinants will grow on the medium containing both the antibiotics. In this case, one antibiotic resistance gene helps in selecting the transformants, whereas the other antibiotic resistance gene gets 'inactivated due to insertion' of alien DNA, and helps in selection of recombinants. Origin of replication (ori) is a

sequence from where replication starts and any piece of DNA when linked to this sequence can be made to replicate within the host cells. This sequence is also responsible for controlling the copy number of the linked DNA. So, if one wants to recover many copies of the target DNA it should be cloned in a vector whose origin support has high copy number.

Which one of the given sites is commonly used by the restriction enzyme?

1. Selectable marker
2. Recognition site
3. Restriction site
4. Vector site

Options 1. 1

2. 2

3. 3

4. 4

Q.48 Read the following paragraph and answer the given questions:

In order to link the alien DNA, the vector needs to have very few, preferably single, recognition sites for the commonly used restriction enzymes. Presence of more than one recognition site in the vector will generate several fragments, which will complicate the gene cloning. In order to link the alien DNA, the vector needs to have very few, preferably single, recognition sites for the commonly used restriction enzymes. Presence of more than one recognition sites within the vector will generate several fragments, which will complicate the gene cloning. The ligation of alien DNA is carried out at a restriction site present in one of the two antibiotic resistance genes. The recombinant plasmids will lose tetracycline resistance due to insertion of foreign DNA but can still be selected out from non-recombinant ones by plating the transformants on tetracycline containing medium.

The recombinants will grow in ampicillin containing medium but not on that containing tetracycline. But, non-recombinants will grow on the medium containing both the antibiotics. In this case, one antibiotic resistance gene helps in selecting the transformants, whereas the other antibiotic resistance gene gets 'inactivated due to insertion' of alien DNA, and helps in selection of recombinants. Origin of replication (ori) is a sequence from where replication starts and any piece of DNA when linked to this sequence can be made to replicate within the host cells. This sequence is also responsible for controlling the copy number of the linked DNA. So, if one wants to recover many copies of the target DNA it should be cloned in a vector whose origin support has high copy number.

In pBR322 'rop' codes for -

1. Translation of the plasmid
2. Transcription of the plasmid
3. Replication of the plasmid
4. Recombination of plasmid

Options 1. 1

2. 2

3. 3

4. 4

Q.49 Read the following paragraph and answer the given questions:

In order to link the alien DNA, the vector needs to have very few, preferably single, recognition sites for the commonly used restriction enzymes. Presence of more than one recognition site in the vector will generate several fragments, which will complicate the gene cloning. In order to link the alien DNA, the vector needs to have very few, preferably single, recognition sites for the commonly used restriction enzymes. Presence of more than one recognition sites within the vector will generate several fragments, which will complicate the gene cloning. The ligation of alien DNA is carried out at a restriction site present in one of the two antibiotic resistance genes. The recombinant plasmids will lose tetracycline resistance due to insertion of foreign DNA but can still be selected out from non-recombinant ones by plating the transformants on tetracycline containing medium.

The recombinants will grow in ampicillin containing medium but not on that containing tetracycline. But, non-recombinants will grow on the medium containing both the antibiotics. In this case, one antibiotic resistance gene helps in selecting the transformants, whereas the other antibiotic resistance gene gets 'inactivated due to insertion' of alien DNA,

and helps in selection of recombinants. Origin of replication (ori) is a sequence from where replication starts and any piece of DNA when linked to this sequence can be made to replicate within the host cells. This sequence is also responsible for controlling the copy number of the linked DNA. So, if one wants to recover many copies of the target DNA it should be cloned in a vector whose origin support has high copy number.

In cloning vector pBR322, the BamH1 site codes for:

1. Ampicillin
2. Chloramphenicol
3. Tetracycline
4. Nuclease

Options 1. 1

2. 2

3. 3

4. 4

Q.50 Read the following paragraph and answer the given questions:

In order to link the alien DNA, the vector needs to have very few, preferably single, recognition sites for the commonly used restriction enzymes. Presence of more than one recognition site in the vector will generate several fragments, which will complicate the gene cloning. In order to link the alien DNA, the vector needs to have very few, preferably single, recognition sites for the commonly used restriction enzymes. Presence of more than one recognition sites within the vector will generate several fragments, which will complicate the gene cloning. The ligation of alien DNA is carried out at a restriction site present in one of the two antibiotic resistance genes. The recombinant plasmids will lose tetracycline resistance due to insertion of foreign DNA but can still be selected out from non-recombinant ones by plating the transformants on tetracycline containing medium.

The recombinants will grow in ampicillin containing medium but not on that containing tetracycline. But, non-recombinants will grow on the medium containing both the antibiotics. In this case, one antibiotic resistance gene helps in selecting the transformants, whereas the other antibiotic resistance gene gets 'inactivated due to insertion' of alien DNA, and helps in selection of recombinants. Origin of replication (ori) is a sequence from where replication starts and any piece of DNA when linked to this sequence can be made to replicate within the host cells. This sequence is also responsible for controlling the copy number of the linked DNA. So, if one wants to recover many copies of the target DNA it should be cloned in a vector whose origin support has high copy number.

Which one of the following is not correct statements about cloning sites of vectors?

1. Presence of more than one restriction site
2. Presence of more than one recognition site in the vector
3. The ligation of alien DNA is carried out at a restriction site
4. The recombinant plasmids will lose tetracycline resistance due to insertion of foreign DNA

Options 1. 1

2. 2

3. 3

4. 4